

SWAPNIL MONDOL

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Academic Profile

Chemical Engineering undergraduate with research experience in surface engineering, biomass processing, and membrane-based separations. Experienced in experimental design, process optimization, and materials characterization, with a focus on developing scalable and sustainable chemical processes for real-world applications.

Education

National Institute of Technology, Rourkela, India

Bachelor of Technology in Chemical Engineering – CGPA: 6.63/10

November 2022 – May 2026

Rourkela, Odisha, India

Jhenaidah Cadet College, Jhenaidah, Bangladesh

Higher Secondary, Science (PCM) – Percentage - 94%

December 2021

Jhenaidah, Bangladesh

Jhenaidah Cadet College, Jhenaidah, Bangladesh

Secondary, Science (PCM) – Percentage - 93%

May 2019

Jhenaidah, Bangladesh

Research Interests

Surface Engineering; Sustainable Materials; Biomass Processing; Membrane-Based Separation; Environmental Systems

Relevant Coursework

- | | | |
|----------------------------------|---------------------------------|------------------------------------|
| • Fluid Dynamics | • Chemical Reaction Engineering | • Chemical Engineering Mathematics |
| • Mass, Heat Transfer Operations | • Chemical Process Calculation | • Computational Fluid Dynamics |
| • Chemical Thermodynamics | • Process Equipment Design | • Separation Processes |

Research Experience

Undergraduate Research Project

Aug 2025 – Present

National Institute of Technology, Rourkela, India

Project: Extraction of Lignin from Biomass via Kraft Process

- Optimized Kraft pulping across 10+ operating conditions by varying 3 parameters—liquor concentration, liquor-to-biomass ratio, and temperature (100–170°C)
- Observed ~ 20% variation in lignin yield, indicating strong sensitivity to operating conditions
- Performed 30+ experimental runs (3+ replicates per condition) to ensure reproducibility and process stability
- Interpreted trends using reaction kinetics and mass transfer, establishing correlations across 10+ datasets for lignin recovery

Winter Research Internship

Dec 2025

Indian Institute of Technology, Kanpur, India

Project: Advanced Desalination Processes — OARO and LSRRO

- Assessed membrane desalination for high-salinity feeds (> 40,000 ppm TDS), comparing 2 advanced configurations (OARO, LSRRO) with conventional RO
- Evaluated performance across 3 metrics—energy consumption, recovery ratio (40–70%), and operating pressure (20–80 bar)
- Identified 15–25% reduction in specific energy consumption from comparative system analysis
- Synthesized insights from 15+ studies to assess efficiency, scalability, and sustainability trends

Summer Research Internship

May 2025 – Jul 2025

National Institute of Technology, Rourkela, India

Project: Superhydrophobic Coatings for Functional Textiles

- Fabricated graphene oxide–metal oxide (ZnO, Fe₂O₃) coatings via sol–gel processing across 5+ deposition conditions
- Achieved water contact angle of 165.8°, corresponding to 35–45% enhancement in hydrophobicity

- Tuned deposition parameters (3 variables) to control surface roughness and surface energy, enabling targeted wettability
- Characterized morphology using 4 techniques and validated durability over 5+ cycles, establishing structure–property relations across 10+ samples

Academic Projects

Superhydrophobic Sponge for Oil–Water Separation Jan 2026 – Present

- Developing porous materials across 3+ configurations for selective oil absorption from wastewater
- Targeting > 10× absorption capacity and contact angles > 150° to enable efficient phase separation
- Assessing reusability via cyclic absorption–desorption over 5–10 cycles
- Evaluating applications in 2 domains—industrial wastewater treatment and oil spill remediation

Energy Storage Devices: Supercapacitors and Batteries Mar 2024 – Apr 2025

- Analyzed 2 charge-storage mechanisms—EDLC and pseudocapacitance—governing electrochemical behavior
- Benchmarked lithium-ion batteries and supercapacitors across 3 metrics—energy density, power density, and cycle life
- Synthesized findings from 20+ studies to evaluate optimization strategies and trade-offs between performance and stability

Research Output

Manuscript Submitted

ZnO–GO–HDTMS Superhydrophobic Nanocomposite Coating on Cotton Fabric: Hierarchical Surface Engineering for Moisture-Resistant Flexible Electronic Protection

- Examines structure–property relationships across 10+ samples linking surface morphology to wettability and durability
- Achieves contact angle ~ 165°, corresponding to 35–45% enhancement in water repellency via sol–gel synthesis

Professional Experience

Program Trainee — Aspire Leadership Program Aug 2024 – Nov 2024

Aspire Institute, Cambridge, Massachusetts, USA

- Applied structured analytical frameworks across 5+ case scenarios within multidisciplinary teams of 4-6 participants
- Completed 10+ modules in ethical leadership, decision-making, and problem-solving methodologies
- Delivered 3+ peer-reviewed presentations, strengthening technical communication and collaborative effectiveness

Achievements/Certifications

- | | |
|------------------------------------|------------------------------|
| • Python Course
Simplilearn | December 2023
Certificate |
| • MATLAB and Simulink
Mathworks | December 2023
Certificate |

Technical Skills

Programming & Simulation: Python, MATLAB, Simulink, DWSIM, ANSYS

Materials Characterization: FESEM, XRD, FTIR, Contact Angle Measurement

Experimental Techniques: Sol–gel synthesis, surface modification

Data & Tools: Microsoft Office , Excel , PowerPoint , Google Sheets

Extracurricular Activities

AIChE Student Chapter May 2023 – Present

Member Portfolio

- Designed 5+ sponsorship and outreach materials for national-level engineering events
- Created 10+ branding assets adopted across digital and print platforms
- Supported engagement initiatives reaching 50+ participants within the chemical engineering community